

Surface Pond Environment Model

LIFEORIG working model specification

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1 Purpose

The surface pond environment is a shallow aqueous reactor exposed to atmosphere, UV radiation, sediment or mineral surfaces, and changing water depth. It is different from a hydrothermal vent and different from volcanic rock pores.

The key physical idea is:

shallow pond $\rightarrow$ water-depth cycles $\rightarrow$ concentration/dilution $\rightarrow$ UV and sediment chemistry

(1)

The first implementation should be a single well-mixed pond. Spatial structure can be added later.

2 Geometry

The first surface pond model is zero-dimensional: one well-mixed shallow water body above a sediment or mineral surface.

```
atmosphere / UV input
  |
  v
+-----+
| shallow pond water      |
| dissolved molecules      |
+-----+
| sediment / mineral floor |
+-----+
```

The geometry is defined by:

$$V(t) = A_{\text{pond}}h(t), \tag{2}$$

where $V(t)$ is water volume, A_{pond} is pond area, and $h(t)$ is water depth.

3 State variables

Symbol	Code name	Meaning
t	<code>time</code>	simulation time

A_{pond}	area	horizontal pond area
$h(t)$	water_depth	instantaneous water depth
$V(t)$	volume	water volume, $A_{\text{pond}}h(t)$
T	temperature	pond water temperature
P	pressure	local pressure
pH	pH	acidity/basicity of pond water
I	ionic_strength	dissolved salt strength
F_{UV}	uv_flux	UV forcing level
A_{sed}	sediment_surface_area	reactive sediment/mineral area
C_i	concentrations[species]	concentration of species i

4 Water-depth cycle

The simplest water-depth cycle is periodic:

$$h(t) = h_0 + \Delta h \cos\left(\frac{2\pi t}{\tau_{\text{cyc}}}\right), \quad (3)$$

with clipping:

$$h(t) \geq h_{\min}. \quad (4)$$

The variables mean:

- h_0 is the mean water depth.
- Δh is the amplitude of drying/refilling.
- τ_{cyc} is the cycle period.
- $h_{\min}$ prevents the pond from becoming mathematically negative.

5 Concentration and dilution

When the pond dries, the same number of molecules occupies less water. The concentration rises. When the pond refills, concentration falls.

If h_{old} is the previous depth and h_{new} is the new depth:

$$C_i^{\text{new}} = C_i^{\text{old}} \frac{h_{\text{old}}}{h_{\text{new}}}. \quad (5)$$

This is the central surface-pond mechanism. It is why shallow ponds can drive prebiotic chemistry: second-order and higher-order reactions speed up strongly when concentrations increase during drying.

6 Chemical update

The minimal concentration equation is:

$$\frac{dC_i}{dt} = R_i(\mathbf{C}, T, \text{pH}, I, F_{\text{UV}}, A_{\text{sed}}) + S_i^{\text{atm}} - k_i^{\text{photo}} F_{\text{UV}} C_i - k_{\text{wash}} C_i - \frac{C_i}{h} \frac{dh}{dt}. \quad (6)$$

The terms mean:

- R_i is production or loss from the chemical reaction network.
- S_i^{atm} is atmospheric input or deposition of species i .
- $k_i^{\text{photo}} F_{\text{UV}} C_i$ is UV photolysis loss.
- $k_{\text{wash}} C_i$ is washout or runoff loss.
- $-(C_i/h)(dh/dt)$ is concentration during drying or dilution during refilling.

7 Code structure

The first implementation is:

SurfacePond

```
state: PondState
concentrations
atmospheric_input
photolysis_rates
water_depth(t)
```

PondState

```
time
water_depth
temperature
pressure
pH
ionic_strength
uv_flux
sediment_surface_area
volume
concentrations
```

8 Comparison with other environments

Environment	Main driver	Main geometry
Surface pond	UV plus drying/refilling concentration	shallow well-mixed water body
Volcanic rock	wet/dry films and rock pores	porous surface rock/interstices
Hydrothermal vent	temperature, pH, redox, and flow gradients	porous chimney wall

9 First implementation target

The first target is not a full ecological model. It is:

1. hold a shallow pond state;
2. update water depth in time;

3. concentrate or dilute chemical species as water depth changes;
4. add atmospheric source terms;
5. apply UV photolysis and washout losses;
6. expose the state to the chemical reaction network.